

Module 8 :: Synthesis of Finite-State Machines

Practice Problems

1. Design a synchronous counter that counts in the following sequence using (i) J-K flip-flop, (ii) T flip-flop, (iii) D flip-flop:
 - a) 0000, 0011, 0110, 1001, 1100, 0111, and then back to 0000.
 - b) 000, 111, 110, 001, 101, 110, 100, 011, and then back to 000.
 - c) 000, 001, 011, 010, 110, 111, 101, 100, and then back to 000.
 - d) 000, 001, 001, 100, 100, 110, and then back to 000.
2. Design a 3-bit synchronous up/down counter. Depending on the value of a control input U/D' , the counter will be counting up or down with every applied clock.
3. It is required to design a serial adder. Draw the state diagram, and implement it using a D flip-flop.
4. It is required to design a sequence detector that takes a serial bitstream X and outputs a serial bitstream Z . The next bit of Z will be 1 every time the bit sequence 1001 is detected at the input. You must consider overlapping patterns in the input stream. Design the circuit using D flip-flops.
5. Repeat (4) for the bit stream 1010.
6. Repeat (4) and (5) for the design using S-R flip-flops.
7. Consider the design of a multiplier using repeated addition. Implement the control path as an FSM, and design it using D flip-flops.
8. Consider the design of a circuit to compute the GCD of two 16-bit unsigned integers. Draw the data path and show the control path specification as an FSM.